

Student Name:

Student ID:

Habib University
Dhanani School of Science and Engineering
Computer Architecture
EE 371 / CS 330 / CE 321 (Fall 2024)

Quiz 4 1

Date: 27th Nov., 2024

Total marks: 20

Obtained Marks:

Total time: 50 mins

Instructions:

1. All questions should be answered **with ink pen**.
2. Smart watches, laptops, and similar electronics are **strictly NOT allowed**.
3. Answer sheets should contain all steps, working, explanations, and assumptions.
4. Write your name and HU ID on all sheets.
5. Turn in your question paper along with your answer sheets.
6. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This quiz assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	<i>Explain</i> the role of ISA in modern processors and instruction encodings and assembly language programming	
CLO 2	<i>Explain</i> the architecture and working of a single cycle processor	
CLO 3	<i>Design</i> the architecture to mitigate issues of a pipelined processor	
CLO 4	<i>Analyze the</i> performance of cache operations	✓

All parts carry equal marks

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Assume we have 64-bit addressing and a 64KiB direct mapped cache having 8 words per block.

a) How many bits do we need for tag, index and word offset?

Sol:

$$2^m = 8 \rightarrow m = 3 \text{ bits (word offset)}$$

$$\text{Total blocks} = (64 \times 1024) / (8 \times 4) = 2048 \text{ blocks} \rightarrow 2^n = 2048 \rightarrow n = 11 \text{ bits (index)}$$

$$\text{Tag} = 64 - n - m - \text{byte_offset} = 64 - 11 - 3 - 2 = 48 \text{ bits}$$

b) Calculate total number of bits needed for the cache.

Sol:

$$\text{Single cache block/line} = \text{Tag bit} + \text{Valid bit} + \text{block size}$$

$$\text{block size} = 8 \times 4 \times 8 = 256 \text{ bits}$$

$$\text{Single cache block} = 48 + 1 + 256 = 305 \text{ bits}$$

$$\text{Total bits} = \text{single cache block} \times \text{total blocks} = 305 \times 2048 = 624640 \text{ bits or } 76.25 \text{ KiB}$$

c) For the following word addresses, find block address and index for the given cache [4 points]

1400

Sol:

$$\text{Block address} = \text{word address} / \text{bytes per block} = 1400 / (8 \times 4) = 43.75 \rightarrow 43$$

$$\text{Index} = \text{Block address} \% \text{ No. of cache blocks} = 43 \% 2048 = 43$$

57600

Sol:

$$\text{Block address} = \text{word address} / \text{bytes per block} = 57600 / (8 \times 4) = 1800$$

$$\text{Index} = \text{Block address} \% \text{ No. of cache blocks} = 1800 \% 2048 = 1800$$

172800

Sol:

$$\text{Block address} = \text{word address} / \text{bytes per block} = 172800 / (8 \times 4) = 5400$$

$$\text{Index} = \text{Block address} \% \text{ No. of cache blocks} = 5400 \% 2048 = 1304$$

1363968

Sol:

$$\text{Block address} = \text{word address} / \text{bytes per block} = 1363968 / (8 \times 4) = 42624$$

$$\text{Index} = \text{Block address} \% \text{ No. of cache blocks} = 42624 \% 2048 = 1664$$

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- d) If blocks per word is kept constant and we convert this cache into a 4-way set associative cache, find number of bits for the tag and index part.

Sol:

We will have $\frac{1}{4}$ of the total cache blocks $\rightarrow$ index = index/4 = 2048/4 = 512

$$2^n = 512 \rightarrow n = 9 \text{ bits (index)}$$

$$\text{Tag} = 64 - n - m - \text{byte_offset} = 64 - 9 - 3 - 2 = 50 \text{ bits}$$

- e) Assume a processor is running at 2GHz. The main memory has an access time of 150 nsec. For this processor, a perfect cache (i.e. no misses) has a base CPI of 1.5. Find the new CPI for this processor if it has a L1 instruction cache with a miss rate of 3% and L1 data cache with a miss rate of 5%. Assume that the data instructions constitute 33% of the total instructions.

Sol:

$$\text{cc time} = 1/2\text{GHz} = 0.5 \text{ nsec}$$

$$\text{miss penalty} \rightarrow 150 \text{ nsec}/.5 = 300 \text{ cc}$$

$$\text{CPI}_{\text{new}} = \text{CPI}_{\text{base}} + \text{CPI}_{\text{cache}}$$

$$\text{CPI}_{\text{cache}} = 3\% * 300 + 5\% * 300 * 33\% = 9 + 4.95 = 13.95$$

$$\text{CPI}_{\text{new}} = 1.5 + 13.95 = 15.45$$